

5 (a)	$\langle P \rangle = \frac{V_{\text{r.m.s.}}^2}{R} = \frac{\left(\frac{V_0}{\sqrt{2}} \right)^2}{R} = \frac{V_0^2}{2R} = \frac{1}{2} P_{\text{max}}$
5(b)(i)	$\omega = \frac{2\pi}{T} = \frac{2\pi}{18 \times 10^{-3}} = 349 \text{ rad s}^{-1}$
5(b)(ii)	$V_{\text{r.m.s.}_1} = \frac{1}{\sqrt{2}} V_0 = \frac{170}{\sqrt{2}} = 120 \text{ V}$
5(c)	Mean square V, $\langle V^2 \rangle = \frac{\text{area bounded by } V^2(t) \text{ in 1 cycle}}{\text{period}}$ $= \frac{((170)^2 \times 9 \text{ ms})}{18 \text{ ms}}$ $\therefore V_{\text{r.m.s.}} = \sqrt{\langle V^2 \rangle} = 120 \text{ V}.$

	The majority of answers did not average the squares of the voltage shown for one cycle.
5(d)(i)1.	A varying e.m.f. source across the primary coil resulted in a varying magnetic field being produced. This varying magnetic field is linked to the secondary coil via the soft iron core. Thus, the magnetic flux linkage of the secondary coil is varying and by Faraday's law, an induced e.m.f. will generated in the secondary coil as the rate of change of magnetic flux linkage is non-zero. The terms 'flux linkage' and 'rate of change of flux linkage' is often omitted when quoting Faraday's law.
5(d)(i)2.	For induced e.m.f. in the primary coil, $E_p = -\frac{dN_p\Phi}{dt} = -N_p \frac{d\Phi}{dt} = V_1$ $E_p = V_1 \text{ for open circuit.}$ $\Rightarrow -\frac{d\Phi}{dt} = \frac{V_1}{N_p}$ For the secondary coil, $E_s = -\frac{dN_s\Phi}{dt} = -N_s \frac{d\Phi}{dt} = V_s \text{ for open circuit too.}$ $\Rightarrow -\frac{d\Phi}{dt} = \frac{V_s}{N_s}$ Since the transformer is ideal, rate of change of magnetic flux is the same. $\frac{V_1}{N_p} = \frac{V_s}{N_s}$ Since $N_s < N_p$, $V_s < V_p$. Students are not to use $\frac{V_1}{N_p} = \frac{V_s}{N_s}$ to explain but to use Faraday's law as required in the question. A significant number of answers gave the reason for the e.m.f. being less for the secondary coil as due to eddy current losses or heat losses in the $15\ \Omega$ resistor, even though the transformer was stated as being ideal and the secondary circuit was shown as an open circuit.
5(d)(ii)	For the secondary coil, $\frac{V_1}{N_p} = \frac{V_s}{N_s} \Rightarrow V_s = N_s \frac{V_1}{N_p} = 20 \frac{120}{500} = 4.8\ \text{V}$ $\Rightarrow I_s = \frac{V_s}{R} = 0.32\ \text{A}$

	$I_P = I_S \frac{N_S}{N_P}$ $\Rightarrow I_P = (0.32) \frac{20}{500} = 0.013 \text{ A}$
6(a)(i)	The period T increase with increase in l but with a decreasing rate.